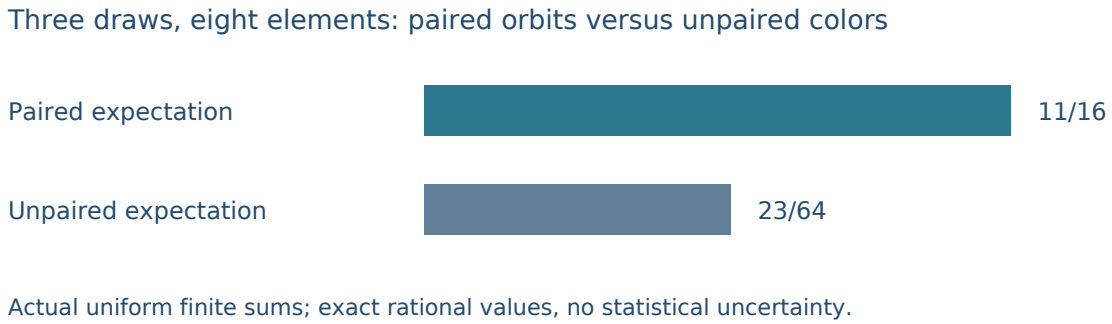
Actual iid sampling: recurrence and repaired bound

Result. Lean derives Thériault's collision recurrence and closed form from actual uniform finite samples, then proves the corrected quadratic bounds directly. The recurrence is a conclusion, not a hypothesis. The original finite lower bound has a source-domain counterfamily with arbitrarily large even pools at three draws. Signed standing unverified. [pcc:unverified]



Exact sampling proof. For every finite nonempty color type of cardinality m and every sample size s, the actual sample type is all functions Fin s→colors, equally weighted. The natural repeat count is s minus the number of distinct sampled colors; support cardinality≤s justifies its rational cast. Functions avoiding a fixed color are equivalent to functions into the complement subtype, so their number is $(m-1)^s$. Summing missing-color indicators and interchanging the two finite sums yields $E=s-m+m(1-1/m)^s$. No probabilistic approximation or recurrence premise enters this derivation. [pcc:unverified]

Actual paired model. An explicit function-to-product equivalence proves uniform product draws preserve the projected color average for every finite nonempty nuisance-label type. With Boolean signs, the pool has n=2m elements, negation flips the sign, has no fixed points, and equal colors characterize equality up to negation. The actual expectation equals the source closed form $(n/2)(1-2/n)^s+s-n/2$ and satisfies $E_0=0$, $E_{s+1}=(1-2/n)E_s+2s/n$. [pcc:unverified]

Direct repair and family. Applying the byte-identical prior recurrence-bound module now gives actual $4s^2/(9n)\le E_s\le s^2/n$ whenever $3\le s\le n/2$. The actual pools Fin(t+4)×Bool have unbounded cardinalities n=2(t+4). At s=3 each satisfies the strict source domain and violates the printed$2s^2/(3n)$ lower bound. Paired eight-element draws give11/16; treating the same number of elements as eight distinct colors gives23/64, a negative control against forgetting the negation quotient. The asymptotic runtime theorem is not refuted by this finite constant error. [pcc:unverified]

Sources and validation. N.Thériault, ASIACRYPT2003,LNCS2894,75–92, DOI10.1007/978-3-540-40061-5_5, relevant definitions and original pp85–86 inspected in the preceding immutable packet. Pinned Mathlib finite-cardinality/subtype/function-equivalence/sum APIs read. Prior source review and recurrence module provenance are retained. Scalar supplement IDC1h939108711fe75407 is reused from its original immutable source, with the same symbolic cubic proof and official32+32-point check; no new native issuance. KB unavailable, bounded primary/formal fallback recorded. [pcc:unverified]

Assurance and boundary. Whole gate 32/0failures/0trusted; Lean4.35.0/Mathlib56a00585… pinned. Complete source binding, custom source/config/artifact pins, failed/successful attempts, SVG and this inspected PDF retained. Only propext/Classical.choice/Quot.sound; no holes, custom axioms, native evaluator, raised limits, global flags or waivers. Exact PCC replay and signed standing are separate. Actual algorithm walks and Jacobian/subgroup distributions are not proved iid; full runtime/rank/invertibility remain open. No full-paper read, corpus promotion, novelty claim, native experiment, paid compute or upload. [pcc:unverified]